ResMed

AirView™ EXCHANGE

Data exchange platform

AVX Data Collection and Usage Best Practices Guide

English

Contents

1 Overview

The information in this best practices guide supplements the *AVX Patient Management Best Practices Guide*.

This document covers best practices for the optimization of data collection and usage APIs so you can design your integration with efficiency in mind. We highly recommend that you review each workflow to see what best practices you can apply to your integration setup.

2 Get session data (live system)

2.1 Workflow overview

The instructions below outline how you can export data from AirView into your live system.

Follow this workflow to retrieve patient therapy data through AVX and display it in your system. You can also build custom reports in your system.

2.2 Workflow tips

- Session date: Refers to the date when the patient underwent a sleep therapy session (put on and took off the mask).
- Received on date: Refers to the date when AirView received data from the device. For example, if an HME uses an SD card to upload 30 days of data to AirView, the received on date for all sessions is the same but the session dates are different.
  Note: We designed our patient data notifications to work with the **Get Data Received On** API, but not the **sessionDate** API (GET <host>/v1/patients/{ecn}/data/sessions/{date}).

You should remember the following when you retrieve data from AirView:

- AirView continues to send data until your integrating application receives a 200 or 400 response code.
- AirView only supports basic authentication when it sends notifications.
- A patient's sleep pattern can update the session date throughout the day (AirView sends a notification when this occurs).
- When you call to get data, AirView returns a complete day summary that refreshes sleep session data in your database.
  Note: We recommend that you save the receipt time in your system. Your application can detect updated sleep session data when you store the receipt time in your database along with the sleep data.
- AirView sends a notification for each sleep session. Your application should wait before it calls to retrieve the sleep data based on the receivedOn date so only one call is made for a batch of sleep sessions.
- AirView queues unsent notifications in case of technical issues. Therefore, your system still receives a complete record of notifications even if there is an outage.

2.3 Required APIs

- **Get Patient List** (If you do not have the patient's ECN stored)
  (GET <host>/v1/orgs/{orgID}/patients)
- **Subscribe to Patient Notifications**
  (POST <host>/v1/subscriptions/patients/{ecn}/data/sessions)
  Note: This API provides you with a subscription Universally Unique Identifier (UUID) that you should save so you know you subscribed to a particular patient for notifications.
- **Get Data Received On**
  (GET <host>/v1/patients/{ecn}/data/sessions?receivedOn={date})

2.4 Instructions

We recommend that you use data notifications to retrieve therapy data from AVX. If you create patients through AVX, we recommend that you store the patient ECN and the subscription for notifications at the same time.

1. Use your list of patient ECNs to call the **Subscribe to Patient Notifications** API.
2. (Optional) If you create patients through AVX, subscribe for patient data notifications once you receive the ECN from the **Create Patient** API.
3. Create an endpoint to retrieve AVX notifications.
4. When you get a new patient data notification from AirView, call the **Get Data Received On** API with the data returned in the resource parameter.

3 Get session data (data warehouse)

3.1 Workflow overview

The instructions below outline how you can export data from AirView into your data warehouse.

Follow this workflow to store patient therapy data in your data warehouse.

3.2 Workflow tips

- Once you have an ECN, you can get all data and reports for a particular patient.
- No individual API can return data for an entire organization; therefore, you should compare ECNs from the **Get Patient List** API with ECNs already in your system and call the **Get Data Received On** API for new ECNs.

  When AirView provides cumulative device session data for a single sleep session over two consecutive days, update data continuously instead of an additive approach. The data elements and the session date remain the same, but the receipt time changes and the data element totals increase and/or the averages change based on the cumulative usage.
- maskOn and maskOff are arrays, but they are independent arrays within the JSON structure. These arrays come in pairs but in some cases one item in the mask on/off pair is missing.
- AirView sends notifications through an HTTP POST via port 443 with the notification content in the body of the request. AVX uses a username, password and basic authentication to connect to your notification handler (must be served over HTTPS). If the IP address or SSL certificate for your notification endpoint changes, notify ResMed in advance.

- Contact your Solutions Consultant or ResMed Program Manager and provide them the static IP (or IP range) for your endpoint so we can whitelist it in our test and production environments. We also require the URL of your endpoint so we can configure the notifications to call over HTTPS and the credentials for basic authorization.

3.3 Required APIs

- **Get Patient List** (If you do not have the patient's ECN stored)
  (GET <host>/v1/orgs/{orgID}/patients)
- **Subscribe to Patient Notifications**
  (POST <host>/v1/subscriptions/patients/{ecn}/data/sessions)
- **Get Data Received On**
  (GET <host>/v1/patients/{ecn}/data/sessions?receivedOn={date})

3.4 Instructions

Whether you use a live system or a data warehouse, we recommend that you use data notifications to retrieve therapy data from AVX.

1. Use your list of patient ECNs to call the **Subscribe to Patient Notifications** API.
2. Create an endpoint to retrieve AVX notifications.
3. When you get a new patient data notification from AirView, call the **Get Data Received On** API with the data returned in the resource parameter.

4 Backfills

4.1 Workflow overview

The instructions below outline how you can complete an initial backfill of patients.

Follow this workflow when you set up your integrating application with AVX for the first time and you have pre-existing patient records you want to backfill.

4.2 Required APIs

Get Patient Data for Session Date Range
(GET <host>/v1/patients/{ecn}/data/sessions?noOfDays={days}&endDate={date})

4.3 Instructions

1. Complete the *Match patients* workflow instructions in the *AVX Patient Management Best Practices Guide*.
2. For each patient, call the **Get Patient Data for Session Date Range** API, where noOfDays is no more than 90.
   Stagger the API call with at least 20 seconds between successive calls. To retrieve all therapy data, you must make repeated calls of 90 days going back to the patient's setup date. If you have a large number of patients in your organization (over 100,000) and you want to process their data concurrently, plan to call the APIs after 5 p.m. Pacific Standard Time on the weekend. If you must run the backfill at another time of the day or week, note that you may receive a 429 (Too Many Requests) response status code from AVX.

5 Differences between APIs used to get session data from AirView

The information below describes the differences between the APIs you can use to retrieve patient session data from AirView to store in your system or data warehouse. These APIs are available in the *AVX API Technical Specification for Data Collection and Usage*.

5.1 /data/sessions/yyyy-MM-dd

Use this API to retrieve data for:

- a single day for a specified date
- ad-hoc requests for individual days.

Do not use it to:

- poll for new data
- get historical data for storage
- get a range of data within a specified date range
- get near real-time data.

5.2 /data/sessions?receivedOn=yyyy-MM-dd

Use this API to:

- retrieve data that was uploaded on a specified date

  Typically, AirView sends a single day of data for the day before the date you specify. However, on occasion, the usage for a patient's device might not have uploaded because of a service outage or the device was not within a cellular coverage area, turned off or on card upload only.

  All available data is uploaded at the same time with the same receipt time. AirView returns all days with a receipt time that matches the **receivedOn** parameter.
- retrieve data when you receive notification of new data from the AVX notification subscription service
- poll for new data on a small number of patients with daily scheduled polling.

  Note: We recommend you do not use this method to poll for new data but use notifications instead.

Do not use it to:

- get historical data for storage
- get data within a specified date range
- get near real-time data in a polling situation.

5.3 /data/sessions?noOfDays={days}&endDate=yyyy-MM-dd

Use this API to:

- retrieve all data within a specified date range
- get historical data for storage
- get a range of data within a specified date range for ad-hoc calculations
- patch missed data (for example, to patch data when the process that populates the data warehouse was off for a few days or weeks).

Do not use it to:

- poll for new data
- get near real-time data.

5.4 About session data

AirView can upload session data multiple times per day. This usually happens when a patient naps during the day. When you upload session data, it replaces the current data and updates the receipt time.

If you use notifications, or try to get near real-time data (not recommended without using notifications), we recommend that you save the receipt time for a session along with the data to determine if the session data was updated.

A patient's card can hold 365 days of therapy summary data. Some patients are card-only or, at times, may have no cellular coverage. Therefore, their upload patterns may indicate that the integrating application missed the new data.

5.5 Example patient session data uploads

The following table shows examples of different patterns for regular uploads, intermittent wireless connection and card-only data uploads for patients with real-world date examples. The information in a cell indicates upload status (Uploaded or Not yet uploaded) and, if applicable, the date the session data was uploaded.

Note: Multiple sessions uploaded in a single day are not shown.

Patient session data uploads

Actual date	2019-01-01	2019-01-02	2019-01-03	2019-01-04	2019-01-05	2019-01-06	2019-01-07	2019-01-08	2019-01-09	2019-01-10
Patient A Regular uploads	Uploaded 2018-12-31	Uploaded 2019-01-01	Uploaded 2019-01-02	Uploaded 2019-01-03	Uploaded 2019-01-04	Uploaded 2019-01-05	Uploaded 2019-01-06	Uploaded 2019-01-07	Uploaded 2019-01-08	Uploaded 2019-01-09
Patient B Intermittent wireless connection	Uploaded 2018-12-31	Uploaded 2019-01-01	Uploaded 2019-01-02	Uploaded 2019-01-03	Uploaded 2019-01-04	Uploaded 2019-01-05	Uploaded 2019-01-06	Uploaded 2019-01-07	Uploaded 2019-01-08	Not Yet Uploaded
Patient C Card-only	Uploaded 2018-12-31	Uploaded 2019-01-01	Uploaded 2019-01-02	Uploaded 2019-01-03	Uploaded 2019-01-04	Uploaded 2019-01-05	Uploaded 2019-01-06	Not Yet Uploaded	Not Yet Uploaded	Not Yet Uploaded

5.5.1 Patient A

This example shows the simpler, usual pattern for Air10™ and newer devices. It's most likely to produce solutions that can lead to gaps in data for patients that don't follow this pattern exactly. It doesn't show delays in data and doesn't prepare you to handle outages for your system or AirView.

5.5.2 Patient B

This example pattern shows the usage that happens at some point in the usage history for a device on wireless coverage. You can see why it's a good idea to implement notifications where possible. You don't need any logic or rolling windows when relying on notifications because you receive notifications when new data is available which indicates the date to supply to the **receivedOn** variant. We recommend that your integrating application also includes a method to retrieve ranges of historical or missing data for patching purposes that is completed once a week or monthly.

5.5.3 Patient C

This is an example of what happens when data is uploaded from a card. When the card is uploaded on January 7, 2019, all days that were not uploaded are uploaded at the same time and stamped with the 2019-01-07 receipt time.

Calling /data/sessions/yyyy-MM-dd for the session dates between 2018-12-31 and 2019-01-06 would not return any data until it was uploaded on 2019-01-07.

Calling /data/session?receivedOn=yyyy-MM-dd for the receivedOn dates between 2019-01-01 and 2019-01-06 return no data. Using the receivedOn date of 2019-01-07 returns all the session data for session dates between 2018-12-31 and 2019-01-06.

If you use notifications, you receive seven notifications at the same time on 2019-01-07. Each notification has a receipt time. For example, "2019-01-07T11:38:21" with slight differences in seconds or minutes. If you implemented the recommended processing delay, you must make only a single API call to retrieve seven days of data in reaction to the notification from AVX.

ResMed

ResMed Ltd
1 Elizabeth Macarthur Drive Bella Vista NSW 2153 Australia

See www.resmed.com for other ResMed locations worldwide.
For patent and other intellectual property information, see www.resmed.com/ip.
 2281472/3 2020-04

ResMed.com